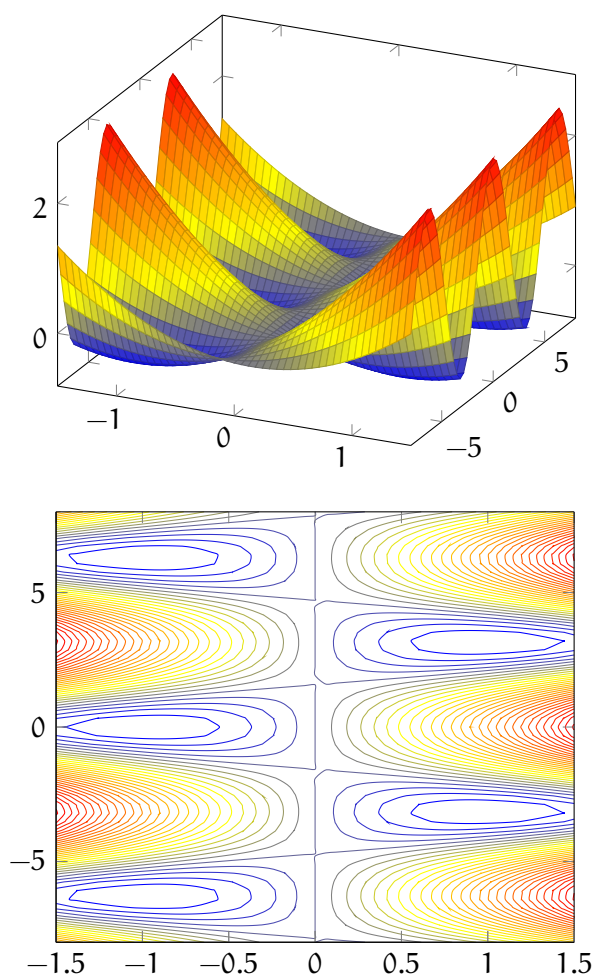


Consider the function

$$f(x_1, x_2) = \frac{1}{2}x_1^2 + x_1 \cos x_2$$

illustrated in the figures below.



Use the optimality conditions to identify the minima of this function.